

LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

Webhook Ingestion & Retry Hub

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Project	Webhook Ingestion & Retry Hub

Structured Jira Evidence and Reflection Report

1. Lab Objective

The objective of this laboratory exercise was to apply Agile project-management practices using Jira. The functional requirements developed in Lab 1 were transformed into a practical Jira backlog consisting of Epics and User Stories. The exercise then focused on prioritization, Fibonacci-based story-point estimation, sprint planning, sprint execution, and interpretation of the Burndown Chart.

The Jira project used for the exercise was the Webhook Ingestion & Retry Hub. The project represents a resilient webhook gateway capable of receiving third-party events, recording delivery information, forwarding events to internal services, retrying failed deliveries, and supporting dead-letter recovery.

2. Backlog and Agile Planning

The backlog was organized into three major Epics: Webhook Integration Management, Webhook Ingestion & Delivery, and Retry & Failure Recovery. These Epics group related functionality and provide a higher-level structure for the User Stories derived from the Lab 1 requirements.

Seven User Stories were created and written from stakeholder perspectives. The stories cover registration of integrations, event ingestion, delivery recording, delivery to a target service, retry handling, dead-letter queue management, and replay of failed webhooks. Priorities were assigned so that the main delivery path received higher priority than recovery functions.

Fibonacci story points were used to represent relative effort and complexity. The assigned estimates were 3, 5, 3, 5, 8, 5, and 3 points respectively, giving a total of 32 story points across the two simulated sprints.

3. Jira Backlog Summary

ID	User Story	Priority	Points	Status
WIRH-4	Register Webhook Integration	High	3	Done
WIRH-5	Ingest Webhook Event	High	5	Done
WIRH-6	Record Webhook Delivery	High	3	Done
WIRH-7	Deliver Webhook to Target	High	5	Done
WIRH-8	Retry Failed Webhook	High	8	Done
WIRH-9	Manage Dead-Letter Queue	Medium	5	Done
WIRH-10	Replay Webhook	Medium	3	Done

The final Jira List view shows all seven User Stories with their priorities and completed status. The three Epics remain visible as the higher-level containers for the project functionality. This final state provides a clear summary of the work created and completed during the laboratory exercise.

4. Sprint Planning and Execution

Two one-week sprints were created for the simulation. Sprint 1 contained WIRH-4 through WIRH-6 and WIRH-9 in the final captured backlog state, while Sprint 2 contained WIRH-7, WIRH-8, and WIRH-10. Each sprint displayed an estimated total of 16 story points in the Jira backlog screenshots.

During execution, work items were progressed through the Scrum-board states To Do, In Progress, and Done. The screenshots document intermediate states as work was moved through the board. This provides evidence that the sprint was simulated as an active workflow rather than only presenting the final Done state.

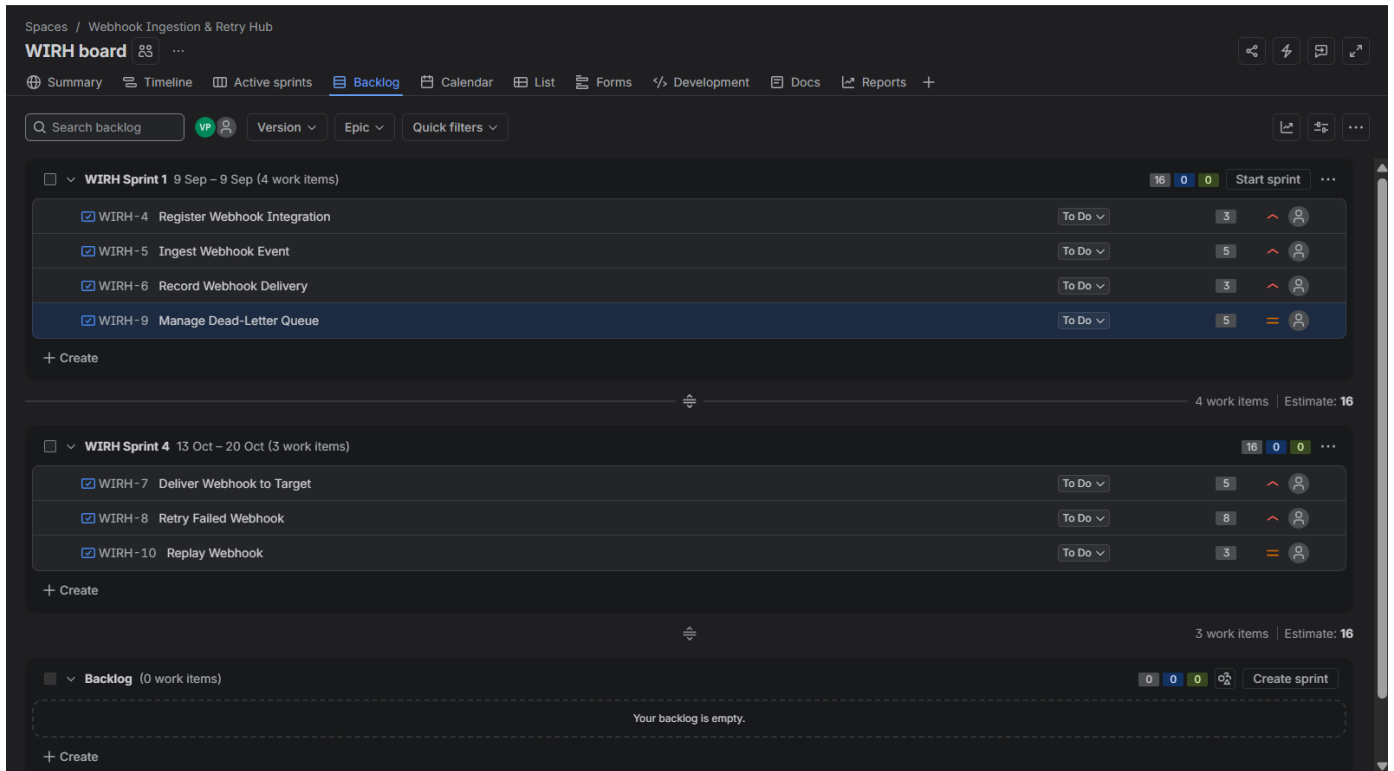
After the sprint simulations were completed, the Burndown Chart was reviewed for each sprint. The charts show the remaining story points falling to zero, indicating completion of the planned work recorded by Jira.

5. Jira Evidence Screenshots

The following screenshots form the main visual evidence for the laboratory submission. The accompanying descriptions explain what each screen demonstrates so that the report can be understood without relying on labels such as 'Screenshot 1' alone.

1. Jira backlog with Sprint 1 and Sprint 2

This screen records the planned sprint structure in Jira. The work items are divided between the two sprint containers, and Jira displays the estimated story-point totals for the selected work.



2. Backlog planning view

This view provides evidence that the User Stories were organized into the sprint plan before execution. It also shows that the backlog was intentionally distributed rather than leaving the stories unplanned.

Q Search

+ Create

See plans

Spaces / Webhook Ingestion & Retry Hub

WIRH board

...

Summary

Timeline

Active sprints

Backlog

Calendar

List

Forms

Development

Docs

Reports

+

Q Search backlog

VP

Version

Epic

Quick filters

...

...

WIRH Sprint 1

9 Sep – 9 Sep (4 work items)

1600

Start sprint

...

WIRH-4

Register Webhook Integration

To Do

3

^

v

...

WIRH-5

Ingest Webhook Event

To Do

5

^

v

...

WIRH-6

Record Webhook Delivery

To Do

3

^

v

...

WIRH-9

Manage Dead-Letter Queue

To Do

5

=

v

...

+ Create

4 work items

Estimate: 16

WIRH Sprint 2

13 Oct – 20 Oct (3 work items)

1600

...

WIRH-7

Deliver Webhook to Target

To Do

5

^

v

...

WIRH-8

Retry Failed Webhook

To Do

8

^

v

...

WIRH-10

Replay Webhook

To Do

3

=

v

...

+ Create

3 work items

Estimate: 16

Backlog

(0 work items)

000

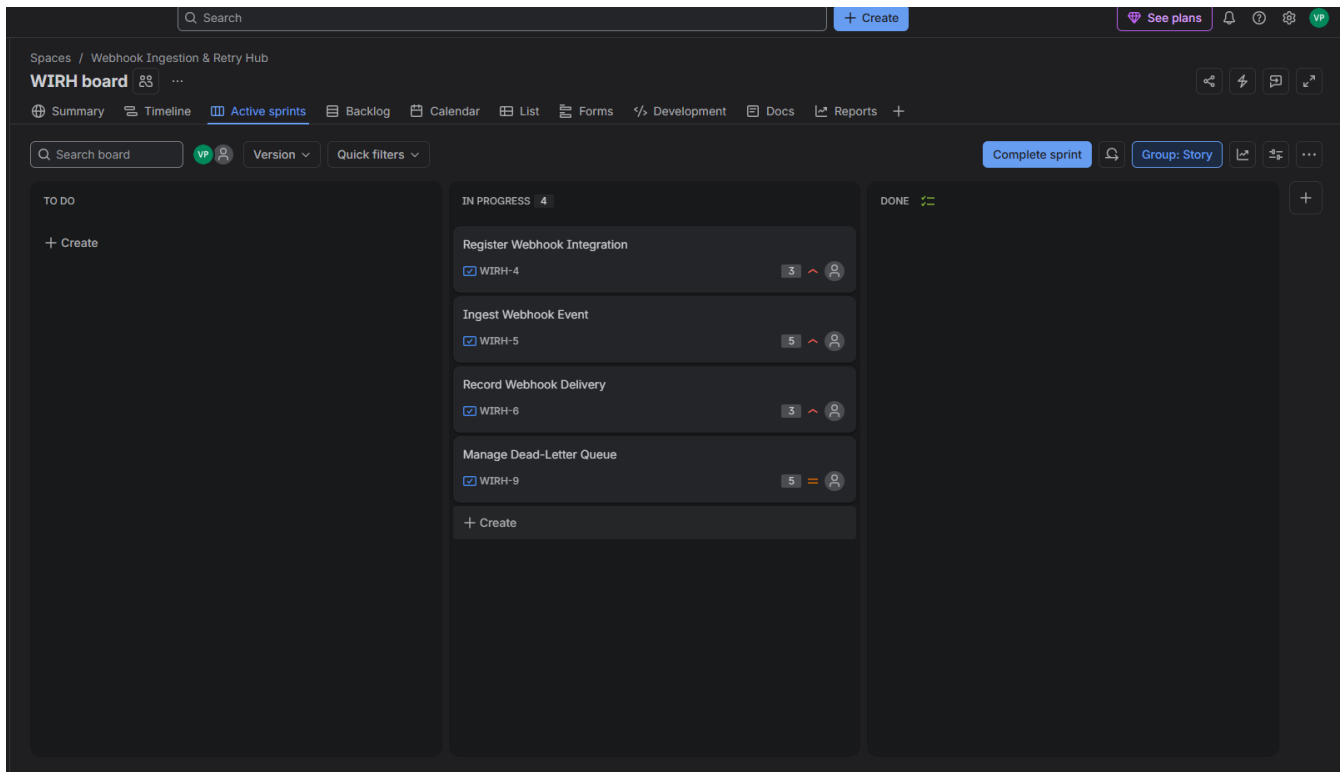
Create sprint

Your backlog is empty.

+ Create

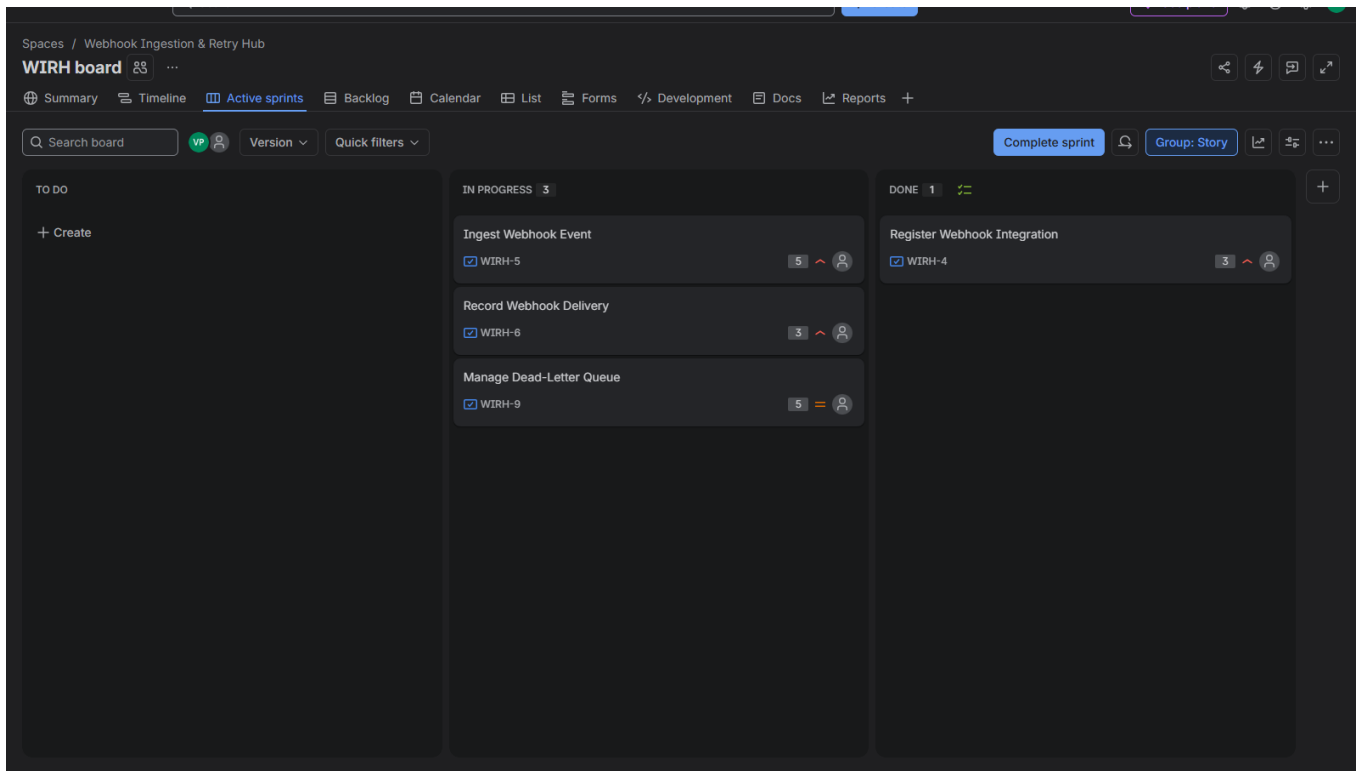
3. Sprint execution — In Progress

The Scrum board shows the selected work moving out of To Do and into In Progress. This represents the active execution stage used in the sprint simulation.



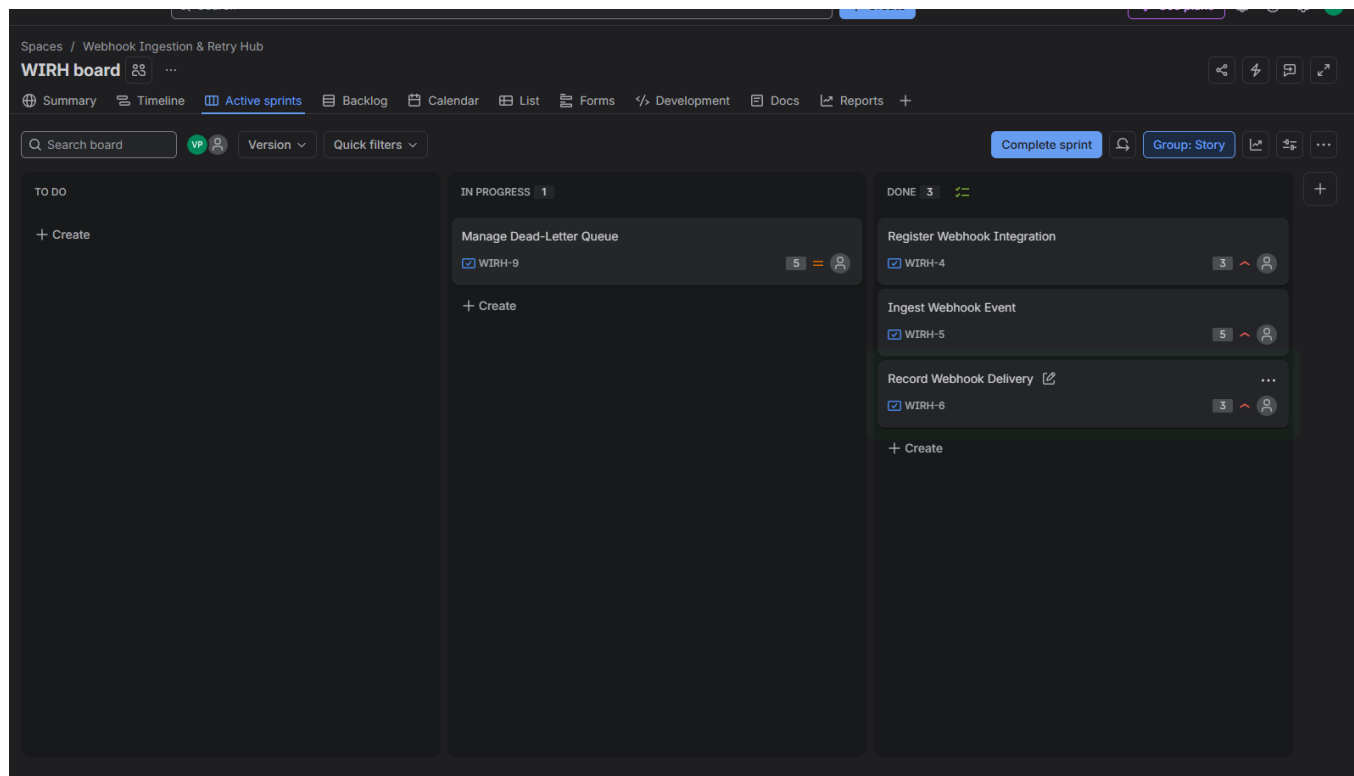
4. Sprint execution — partial completion

This screen demonstrates that completed items can remain in Done while other selected stories continue through In Progress. It provides evidence of incremental progress rather than a single final-state update.



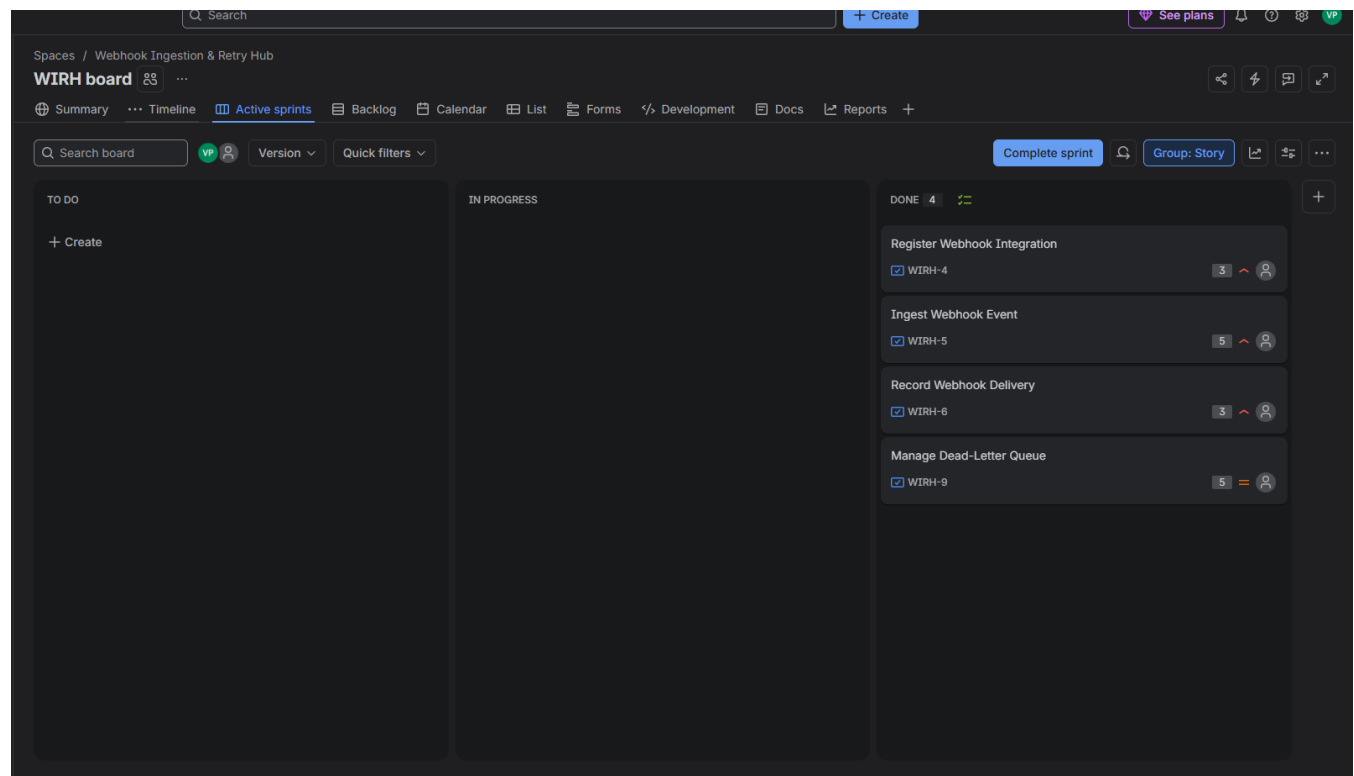
5. Sprint execution — continued progress

Additional stories are shown in progress while earlier work remains completed. The board therefore records the sequence of progress during the simulated sprint.



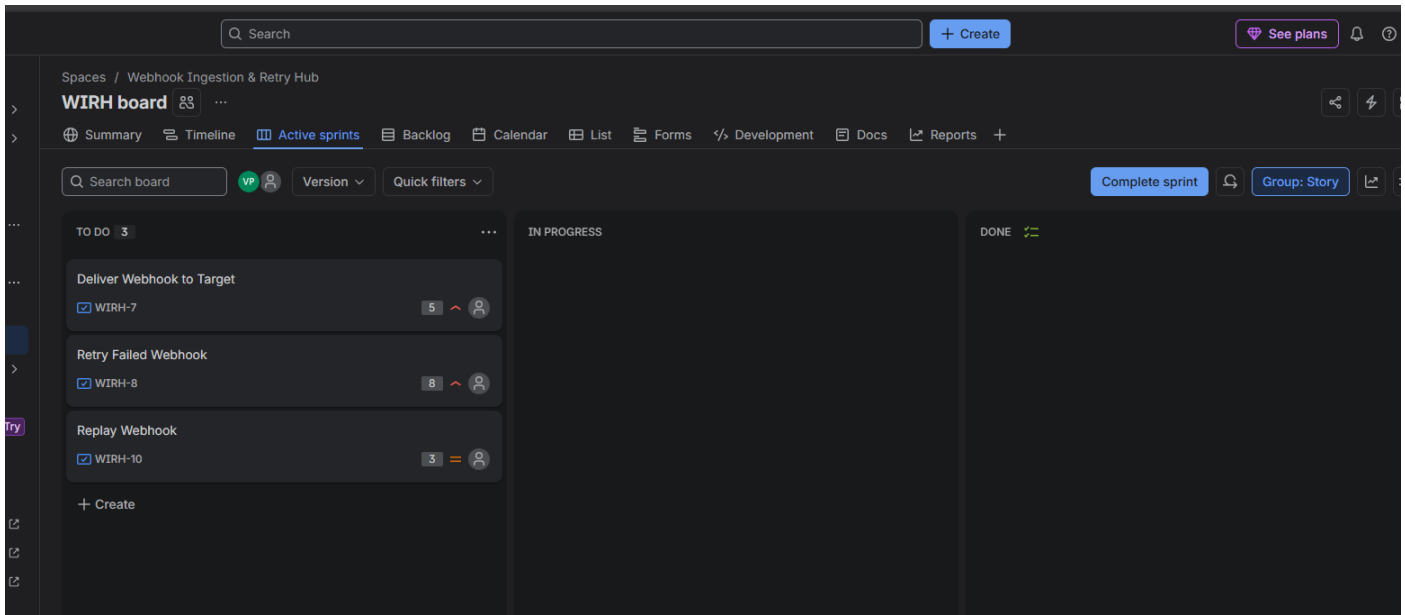
6. Sprint execution — completed work

The selected Sprint 1 work has reached Done. This confirms that the planned work for the first sprint was completed through the Scrum workflow.



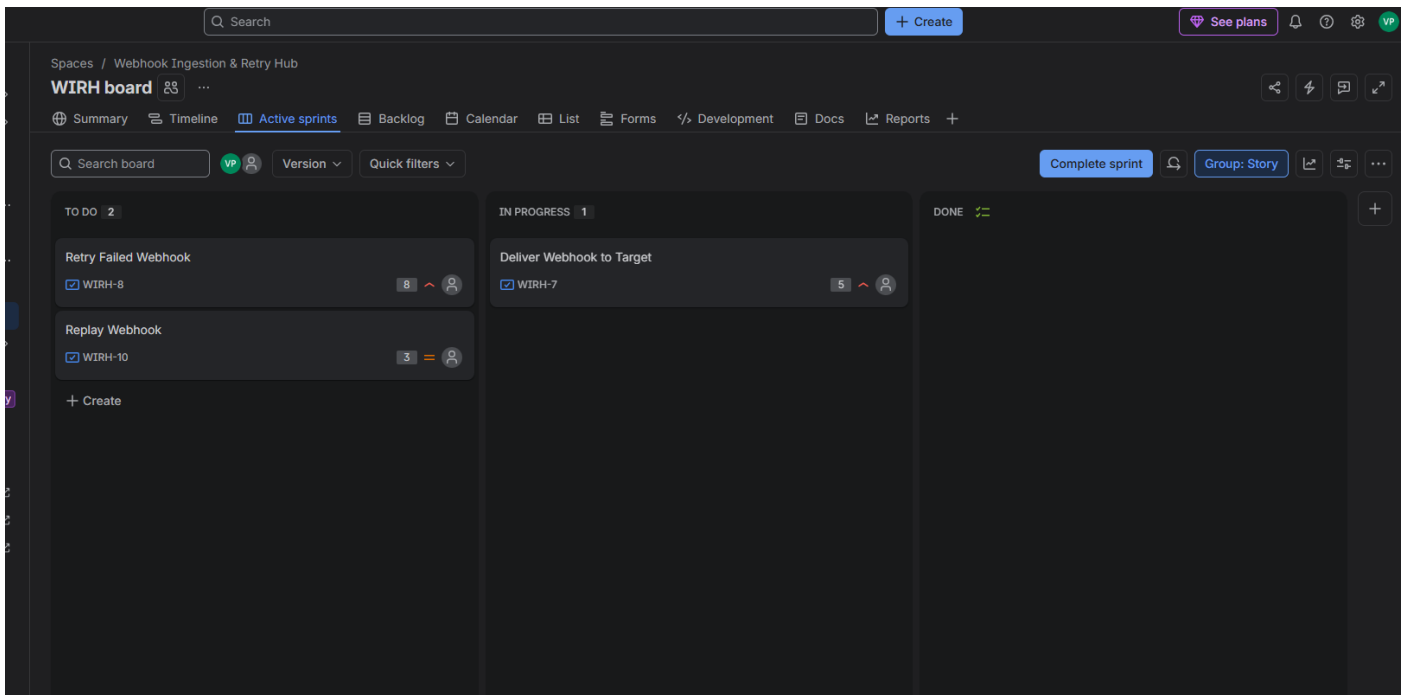
7. Sprint 2 execution — initial progress

The second sprint begins with its selected stories in the To Do state. This provides evidence that the second sprint was separately planned and executed.



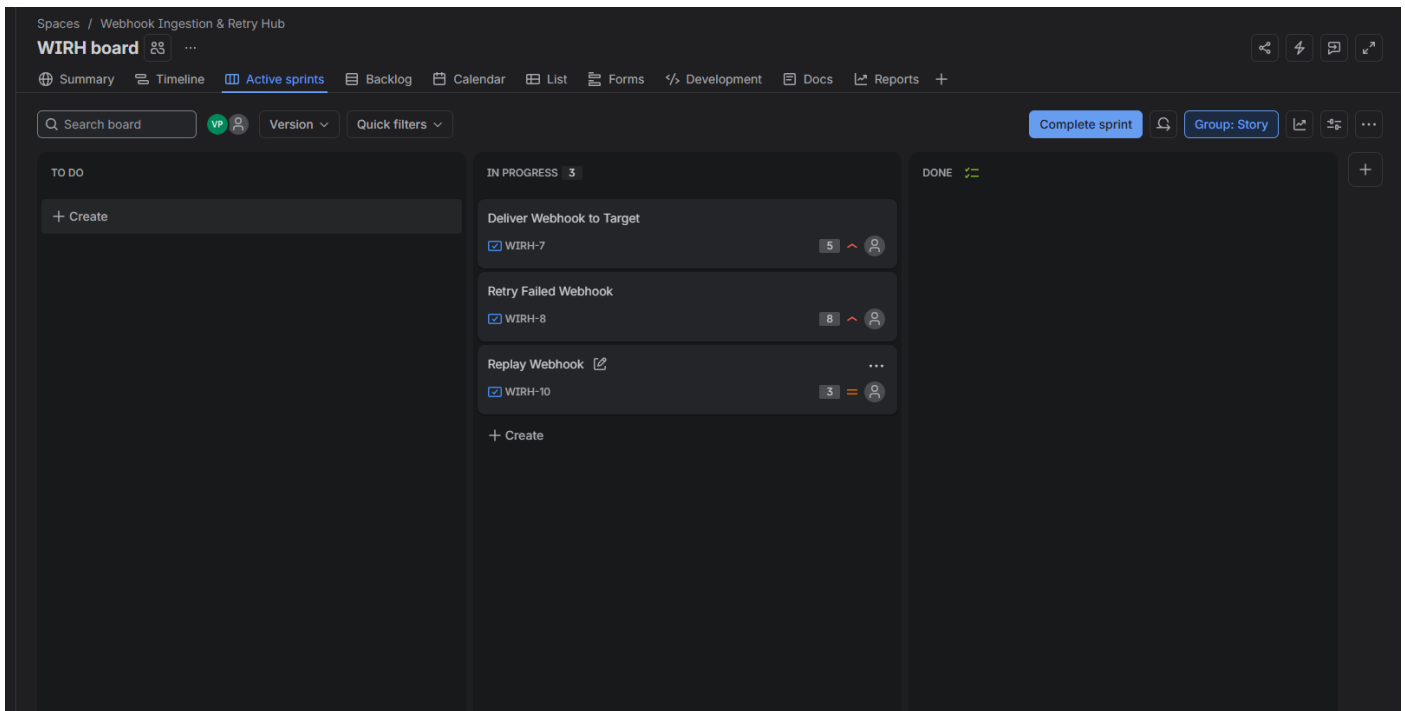
8. Sprint 2 execution — work in progress

Sprint 2 work is being moved into In Progress. The screen demonstrates continued use of the same Scrum workflow for the second time-boxed sprint.



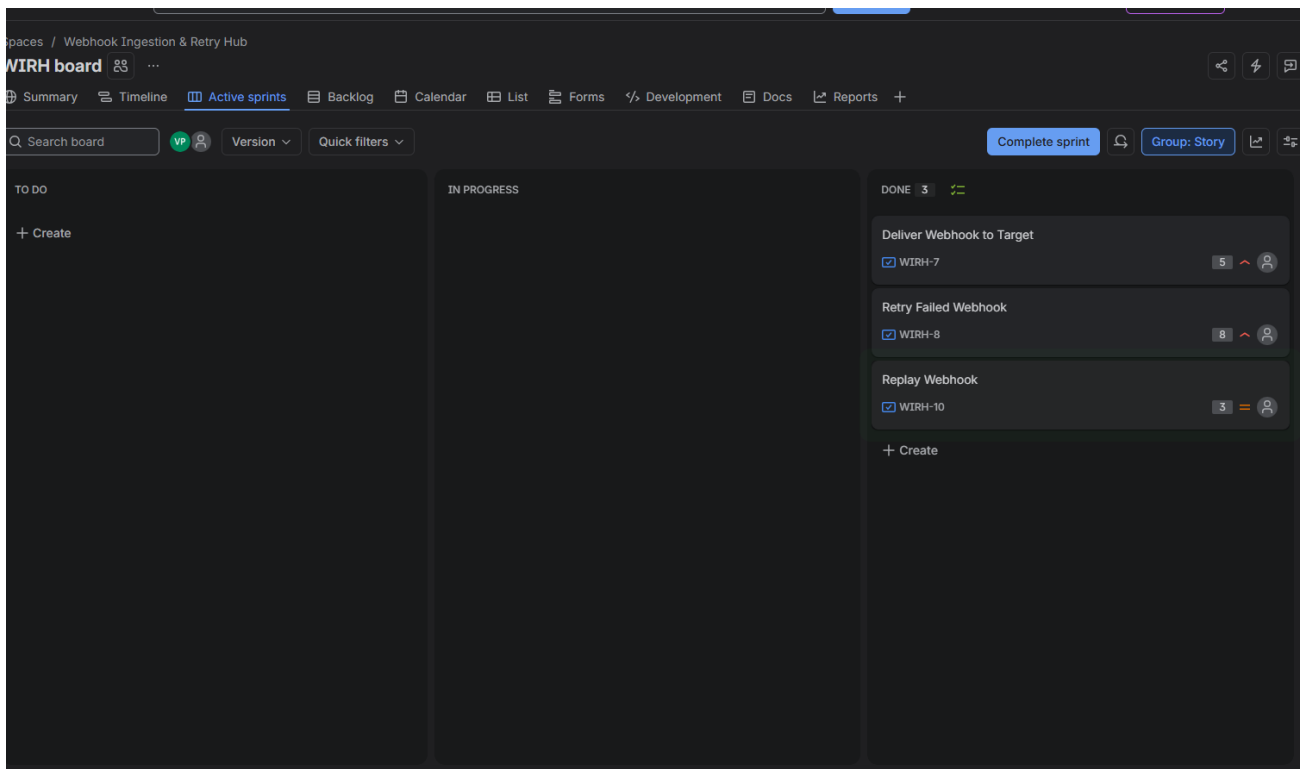
9. Sprint 2 execution — completion

The second sprint reaches its completed state, with the sprint stories progressing toward Done as required by the simulation.



10. Active Sprint board evidence

This compact board view provides additional evidence of the Jira sprint execution environment, including the Scrum columns and completed work items.



11. Sprint 1 Burndown Chart

The Sprint 1 Burndown Chart shows remaining work decreasing from 16 story points to zero. The stepped red line reflects the points being completed during the simulation.



12. Sprint 2 Burndown Chart

The Sprint 2 Burndown Chart also shows remaining work decreasing from 16 story points to zero. The chart provides evidence that the planned sprint workload was completed.



13. Final Jira List view

The final list confirms the seven User Stories and three Epics in the project. The stories show their assigned priorities and completed status, providing a concise final project record.

Spaces / Webhook Ingestion & Retry Hub

WIRH board

Summary

Timeline

Active sprints

Backlog

Calendar

List

Forms

Development

Docs

Reports

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Search work

VP

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Group

	Work	Assignee	Reporter	Priority	Status	
☐	☒ WIRH-4 Register Webhook Integration	☐ Unassigned	☒ VP Vrushant K P	☒ High	☒ Done	
☐	☒ WIRH-5 Ingest Webhook Event	☐ Unassigned	☒ VP Vrushant K P	☒ High	☒ Done	
☐	☒ WIRH-6 Record Webhook Delivery	☐ Unassigned	☒ VP Vrushant K P	☒ High	☒ Done	
☐	☒ WIRH-9 Manage Dead-Letter Queue	☐ Unassigned	☒ VP Vrushant K P	☒ Medium	☒ Done	
☐	☒ WIRH-7 Deliver Webhook to Target	☐ Unassigned	☒ VP Vrushant K P	☒ High	☒ Done	
☐	☒ WIRH-8 Retry Failed Webhook	☐ Unassigned	☒ VP Vrushant K P	☒ High	☒ Done	
☐	☒ WIRH-10 Replay Webhook	☐ Unassigned	☒ VP Vrushant K P	☒ Medium	☒ Done	
☐	☒ WIRH-1 Webhook Integration Management	☐ Unassigned	☒ VP Vrushant K P	☒ Medium	☐ To Do	
☐	☒ WIRH-2 Webhook Ingestion & Delivery	☐ Unassigned	☒ VP Vrushant K P	☒ Medium	☐ To Do	
☐	☒ WIRH-3 Retry & Failure Recovery	☐ Unassigned	☒ VP Vrushant K P	☒ Medium	☐ To Do	

+ Create

10 of 10

6. Sprint and Burndown Analysis

The sprint execution demonstrates the basic Agile feedback cycle. Work begins in To Do, selected stories are moved into In Progress as they are worked on, and completed stories are moved into Done. Recording these state changes in Jira makes sprint progress visible to the team and provides a basis for the Burndown Chart.

For Sprint 1, Jira records an initial workload of 16 story points and the Burndown Chart reaches zero remaining points. Sprint 2 also begins with 16 story points and reaches zero. The step-like reductions in the red remaining-values line indicate that story points were removed from the remaining workload as stories were completed.

The charts also illustrate why completing work incrementally is useful. A team can compare the actual remaining-work line with the guideline and identify whether progress is ahead of, near to, or behind the expected pace. In this simulation, the final state reaches zero.

7. Reflection

7.1 Did your estimations reflect the actual effort?

The Fibonacci estimates provided a useful relative measure of effort and complexity for planning the simulated work. The stories ranged from 3 to 8 points, allowing smaller configuration and recovery activities to be distinguished from the more complex retry functionality. Since the exercise was a simulation rather than a real software-development cycle, there is not enough real implementation-time data to prove that the point values would match production effort exactly. However, the estimates were sufficient to organize the backlog and divide the work across the two sprints.

7.2 Was your backlog well-prioritized?

Yes. The backlog prioritization placed the main webhook-processing path at High priority. Registration, ingestion, delivery recording, target delivery, and retry handling were treated as the most important functions because they directly support successful event processing. Dead-letter queue management and replay were assigned Medium priority because they are primarily recovery-oriented functions that become important after the core delivery path is established.

7.3 How did your simulated sprint align with your plan?

The sprint simulation followed the planned Agile workflow. The selected stories were placed into time-boxed one-week sprints, the sprints were started in Jira, and the stories were progressed through To Do, In Progress, and Done. Two sprints were completed. The screenshots capture both intermediate board states and final completion, showing that the plan was translated into an observable Jira workflow.

7.4 What insights did the burndown chart give about your team's capacity?

The Burndown Charts show each sprint decreasing from 16 remaining story points to zero. This indicates that all planned story points in the simulated sprints were completed. The charts also show that the work was completed in batches rather than at a perfectly uniform rate. In an actual project, several completed sprints would provide more reliable data for estimating sustainable team capacity and improving future sprint commitments.

8. Conclusion

Lab 2 demonstrated the use of Jira for Agile backlog creation, prioritization, estimation, sprint planning, sprint execution, and progress analysis. The Webhook Ingestion & Retry Hub was represented through three Epics and seven User Stories derived from the Lab 1 requirements. The stories were assigned priorities and Fibonacci estimates, organized into two one-week sprint simulations, and progressed through the Scrum workflow.

The final Jira evidence and Burndown Charts provide a complete record of the exercise. The completion of the planned work in both simulated sprints and the reduction of remaining story points to zero demonstrate the intended learning outcomes of the laboratory.

9. Reference

Lab_2_Jira_Student_Handout.pdf — PES University Software Engineering Lab 2 handout. The report structure, sprint workflow, Burndown Chart discussion, and reflection questions are based on the instructions in the supplied handout.